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Seizure Recurrence After a First Unprovoked Seizure in Childhood: A Prospective Study

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Summary: *Purpose:* To study the risk of recurrence after a first unprovoked seizure in childhood.

Methods: All consecutive patients aged less than 14 years with one or more unprovoked seizures who were attended between January 1, 1987, and June 1, 1996, were included in a prospective study. Clinical features of patients attended after a first seizure and those attended after two or more seizures were compared. Recurrence risk in both groups was estimated by Kaplan-Meier curves. Univariate and multivariate analyses of the potential predictors of recurrence risk were performed for the group of patients attended after a first seizure using the Cox proportional hazards model.

Results: Included in the study were 217 children. Kaplan-Meier estimate of recurrence risk was 64% at 5 years, when only patients being attended after a first epileptic seizure were included, compared with 74% when all patients were included. Significant differences in several clinical features were found between patients attended after a first seizure and those attended after two or more seizures. Univariate and multivariate analyses showed that in the overall cohort of patients attended after a first seizure, a symptomatic etiology increased the risk of recurrence, whereas a patient age of 3 to 10 years decreased this risk. In particular, the recurrence risk was 96% at 2 years for symptomatic seizures, compared with 46% for idiopathic/cryptogenic seizures. In the group of patients with idiopathic/cryptogenic seizures, an abnormal electroencephalogram and the occurrence of seizures during sleep increased the recurrence risk, whereas a patient aged 3 to 10 years reduced it. In the group of patients with symptomatic etiology, univariate analysis revealed that there was a lower recurrence risk for patients aged 3 to 10 years. This last finding was not maintained, however, in multivariate analysis.

Conclusions: The recurrence risk depends on the inclusion criteria for enrolling patients. Several factors enable us to predict the recurrence risk after a first unprovoked seizure; the most important of these factors is the etiology of the seizures.
Key Words: Seizure—Epilepsy—Prognosis—Epidemiology—Recurrence.

Knowledge of the risk of recurrence after a first unprovoked seizure, as well as the factors affecting it, is essential when considering the convenience of initiating antiepileptic treatment. However, the rate of recurrence after a first unprovoked seizure varies widely, ranging from 27% to 71%, according to different studies (1–14). The great disparity in the results of different studies is mainly due to methodological differences (15,16). Although today most methodological questions have been agreed on, studies have been published recently that disagree about patient inclusion criteria: some include only those presenting after a first seizure, whereas others include all patients with unprovoked seizures, even if they have a history of more than one seizure when first attended (8,14). Each of these methods could, however, produce the analysis of a different group of patients. Many of these studies are not specifically aimed at examining seizures in childhood, which present important differences from those seizures that occur at later ages (17). For these reasons, we studied the recurrence risk after an initial seizure in childhood using both methodologies and the possible clinical differences between patients attended after a first seizure and those attended after two or more seizures.

METHODS

Cohort selection

The Severo Ochoa Hospital covers the population of the Leganés and Fuenlabrada area, south of Madrid,

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Spain, and provides specialized attention in pediatric neurology. Standard practice of general practitioners and pediatricians in the coverage area is to refer patients with a possible diagnosis of epileptic seizure to this center. All consecutive patients under the age of 14 years with definite unprovoked seizures who were attended between January 1, 1987, and June 1, 1996, were included in a prospective study. Neonatal convulsions and patients already receiving antiepileptic treatment or who had been examined at other centers were excluded.

Definitions and classification criteria

Seizures were considered unprovoked when they occurred in the absence of any known proximate precipitant (9,18–20). More than one seizure within a period of 24 hours was considered to be a single seizure (multiple seizure). Status epilepticus was defined as a seizure lasting more than 30 minutes or as a series of seizures without regaining consciousness between them lasting more than 30 minutes. Seizures were classified according to their etiology as idiopathic, cryptogenic, and symptomatic, following the International League Against Epilepsy (ILAE) criteria (21) and using all the data available when the patient was first examined. The type of seizure was classified following the ILAE international classification of epileptic seizures (22). Electroencephalograms (EEGs) were classified as normal or abnormal, the latter category including both epileptiform (focal or generalized spikes or waves) and nonepileptiform (focal or generalized slowing) abnormalities. A family history of unprovoked seizures was defined as seizures occurring in first-degree relatives.

Initial evaluation and patient follow-up

Patients were examined within a maximum interval of 48 hours. Family and medical history were recorded, a physical and neurological examination was performed, and an EEG was obtained for every patient. All EEGs were performed on a 10- or 20-channel machine using the 10–20 electrode system. All records were interictal, and almost all were obtained while the child was awake. Recording time was at least 30 minutes, including periods with eyes open and closed, and standard stimulation procedures: photic stimulation in every case and hyperventilation in children capable of collaborating. All records were examined by the same person, who was unaware of the clinical data and evolution of the patients.

Computed tomography or magnetic resonance imaging was performed at least in the cases with abnormal findings in the neurological examination, partial seizures, focal abnormalities on the EEG (except in the case of benign childhood epilepsy with centro-temporal spikes), or West syndrome. After the initial visit, patients were followed up by means of a personal interview at maximum intervals of 6 months, if the seizures recurred, or whenever it was considered clinically necessary. Patients were followed from their entry into the study until June 1, 1996 (i.e., the follow-up period lasted as long as that of patients entering the study).

General characteristics of the sample are shown in Tables 1 and 2.

Analysis

Overall recurrence risk

With respect to patient inclusion criteria, two methodologies were used:

1. *First-seizure method* (15), including only those patients who were attended after their first unprovoked seizure and excluding those with any history of unprovoked seizures. Patients were enrolled in the study on the date of the first seizure. The event being studied was the recurrence of seizures.
2. *Recent-onset epilepsy method* (15), including all patients presenting with unprovoked seizures, irrespective of whether one or more seizures had occurred before the first examination. Patients were enrolled in the study on the date of the first seizure. The event studied was the recurrence of seizures, which frequently occurred before the first examination.

In both cases, the recurrence risk after an initial seizure was calculated by Kaplan-Meier survival curves.

TABLE 1. *General characteristics of the sample*

	Seizures before first visit			
	1		>1	
	n	%	n	%
Etiology				
Symptomatic	25	19	22	26
Idiopathic	14	10	34	40
Cryptogenic	94	71	28	33
Age				
0–3 years	36	27	17	20
3–10 years	66	50	41	49
10–14 years	31	23	26	31
Multiple seizures	24	18	31	37
Status epilepticus	13	10	—	—
Todd's paresis	5	4	1	1
Abnormal EEG	31	23	38	45
Seizures while asleep	44	33	23	27
Seizure type				
SPS	9	7	17	20
CPS	22	16	9	11
PSG	39	29	16	19
GTC	60	45	15	18
Other	3	2	27	32
Prior febrile seizures	26	19	5	6
Prior neonatal seizures	5	4	1	1
Family history of seizures	14	10	2	2
First seizure treated	21	16	—	—
Total	133		84	

SPS, simple partial seizures; CPS, complex partial seizures; PSG, partial secondarily generalized seizures; GTC, generalized tonic-clonic seizures.

TABLE 2. *More detailed distribution by age, etiology, and sex of patients with only one seizure before first visit*

	0–3 Years		3–10 Years		10–14 Years		Total	
	n	%	n	%	n	%	n	%
Symptomatic	12	48	9	36	4	16	25	100
Idiopathic/cryptogenic	27	25	57	53	24	22	108	100
Total	39	29	66	50	28	21	133	100

Cases were considered censored if, by the end of the study, the event under observation had not occurred or if contact with the patient was lost.

We also conducted a comparison of the clinical features of the patients who were attended after their first seizure (included in the analysis with both methods) and those attended after two or more seizures (only included in the recent-onset epilepsy method) using the χ^2 or Fisher exact test.

Predictors of recurrence

Potential predictors of recurrence risk were analyzed only for the first-seizure method, because of its obviously greater practical interest. Univariate and multivariate analyses of the potential predictors were performed using the Cox proportional hazards model. When appropriate, the log-rank test was used to compare survival curves. Calculations were performed by means of SPSS for Windows, version 6.1.2, statistical software. The level of statistical significance was established at $p < 0.05$.

RESULTS

Included in the study were 217 patients, 133 (61%) males and 84 (38%) females. The mean age at onset of the first seizure was 7.4 years (±4.05). The average follow-up period was 3.8 years (range, 0.1–9.4 years), with a standard deviation of 2.4 years. Eighty-seven percent of patients were followed for >1 year, 71% for >2 years, 59% for >3 years, 47% for >4 years, 33% for >5 years, and 22% for >6 years. Contact was lost with 3 children after a follow-up period of 0.4 to 0.7 years. One hundred thirty-three patients (61%) had had a single seizure before being first attended, and 84 (39%) had had two or more.

Although it is not our standard practice to start antiepileptic treatment after the first seizure, on occasion it is difficult to avoid this in certain clinical situations. Of the 133 patients who were attended after an initial seizure, 21 (16%) received antiepileptic treatment on the recommendation of their consulting physician or the express wishes of their parents, after being informed of the risk of recurrence of seizures. These patients were also included in the study.

Overall recurrence risk

First-seizure method

We examined 133 patients who had had a single seizure when first attended. The Kaplan-Meier estimate of recurrence risk was 37% (±4.2), 45% (±4.4), 57% (±4.6), 57% (±4.6), 61% (±4.7), and 64% (±4.9) at 6 months, 1 year, 2 years, 3 years, 4 years, and 5 years, respectively (in parentheses, SE) (Fig. 1A). The recurrence risk was greater during the first few months after the first seizure: 18 (24%) of the 75 recurrences occurred during the first month, 46 (61%) during the first 6 months, 58 (77%) within the first year, and 70 (93%) within the first 2 years.

Recent-onset epilepsy method

Two hundred seventeen patients were attended after suffering one or more seizures. The Kaplan-Meier estimate of recurrence risk was 58% (±3.4), 65% (±3.3), 74% (±3.1), 74% (±3.1), 76% (±3.1), and 79% (±3.2) at 6 months, 1 year, 2 years, 3 years, 4 years, and 5 years, respectively (in parentheses, SE) (Fig. 1B). The recurrence risk was greater during the first few months after the first seizure: 68 (43%) of the 159 recurrences occurred during the first month, 125 (79%) during the first 6 months, 138 (87%) within the first year, and 154 (97%) within the first 2 years.

Comparison between the clinical features of patients attended after a first seizure and those attended after two or more seizures

Table 3 illustrates the clinical features for which statistically significant differences between the two groups were observed. For other features listed in Table 1 there were no significant differences. Patients with absences (typical or atypical), myoclonic seizures, atonic seizures, and infantile spasms in all cases were attended after two or more seizures, but the differences were still significant even after this group of patients was excluded from the comparison.

Predictors of recurrence

Overall cohort

Univariate analyses using the Cox proportional hazards model showed that the factors predictive of a significant increase in recurrence risk were a symptomatic

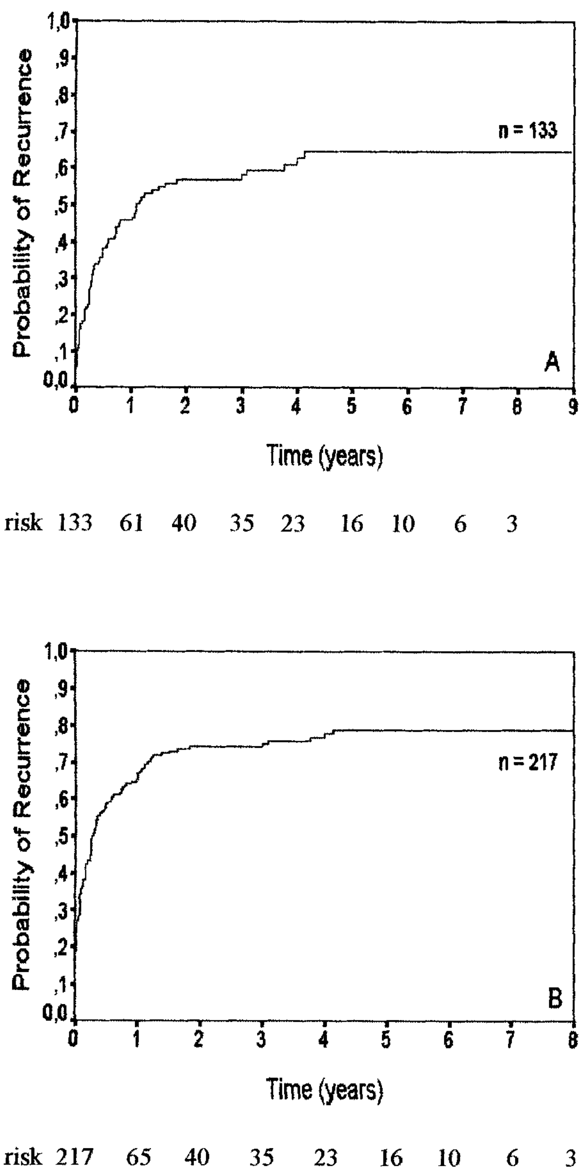


FIG. 1. Risk of seizure recurrence after a first unprovoked seizure. Kaplan-Meier curves. **A:** First-seizure method. **B:** Recent-onset epilepsy method.

etiology and the occurrence of the first seizure during sleep. The presence of an abnormal EEG was almost significant. In contrast, an age between 3 and 10 years predicted a significantly lower recurrence risk (Table 4). Individual risk factors are described in greater detail below.

Etiology of seizures: Patients with a symptomatic first seizure had a greater risk of recurrence than those with an idiopathic first seizure, who in turn had a greater risk than patients with a cryptogenic first seizure. The estimated recurrence risk was 36% and 43% at 1 and 2 years, respectively, for the cryptogenic group, compared with 50% and 67% for the idiopathic group and 76% and 96% for the symptomatic group. The difference was statistically significant both for comparison of the idiopathic and cryptogenic groups ($p = 0.0229$), the symptomatic and cryptogenic groups ($p = 0.0000$), and the idiopathic and symptomatic groups ($p = 0.0197$) (Fig. 2A).

It must be noted that of the 25 patients in the symptomatic etiology group, 15 presented a nonprogressive mental retardation, 7 had cerebral palsy, and only 3 had other causes of epilepsy. To facilitate comparisons with other studies that classified patients into just two etiological groups, we analyzed our results after combining the idiopathic and cryptogenic groups. Patients with a symptomatic first seizure had a higher risk of recurrence than patients with an idiopathic/cryptogenic seizure. The estimated recurrence risk was 76% and 96% for the symptomatic group and 38% and 46% for the idiopathic/cryptogenic group, at 1 and 2 years, respectively (Fig. 2B).

Age at first seizure: Analysis of the recurrence risk according to patient age revealed a lower recurrence risk in children aged 3 to 10 years. Because only a small number of patients were in each age category, we grouped the patients aged between 3 and 10 years to compare them with patients of other ages. Children aged 3 to 10 years had a lower recurrence risk than those of any other age. The Kaplan-Meier recurrence risk was 31% and 44% for the former group, in contrast to 60% and 70% for the latter group, at 1 and 2 years, respectively.

First EEG: Patients with an abnormal first EEG had a higher recurrence risk than those with a normal first EEG. The difference was almost significant.

Sleep state at time of first seizure: Patients whose first seizure occurred during sleep had a higher recurrence risk than those whose first seizure occurred while they were awake. For the former group, the risk was 59% and 75%, compared with 39% and 49% for the latter

TABLE 3. *Comparison between clinical features of patients attended after a first seizure and those attended after two or more seizures*

	Seizures before first visit					
	1		≥2			
	n	%	n	%	P_1	P_2
Etiology						
Idiopathic	14	10	34	40		
Cryptogenic	94	71	28	33	0.00000	0.00048
Symptomatic	25	19	22	26		
Seizure type						
SPS	9	7	17	20	0.003	—
GTC	60	45	15	18	0.00004	—
Abnormal EEG	31	23	38	45	0.0007	0.04
Prior febrile seizures	26	19	5	6	0.005	0.02
Status epilepticus	13	10	0	0	0.002	0.01

SPS, simple partial seizure; GTC, generalized tonic-clonic seizure; P_1, comparison including absences, myoclonic seizures, atonic seizures, and infantile spasms; P_2, comparison excluding these types of seizures (2 × 3 contingency table for etiology and 2 × 2 contingency tables for the other clinical features).

TABLE 4. *Predictors of recurrence risk after a first unprovoked seizure*

	Overall cohort			Idiopathic/cryptogenic			Symptomatic		
Risk factor	RR	95% CI	p	RR	95% CI	p	RR	95% CI	p
Symptomatic etiology	3.8	2.3–6.2	0.000						
Age 3–10 years	0.5	0.3–0.8	0.0020	0.5	0.3–0.9	0.0293	0.4	0.2–0.9	0.0469
Multiple seizures	1.1	0.6–2	0.6692	1	0.5–2	0.9251	1.4	0.6–3.7	0.4559
Status epilepticus	0.6	0.3–1.5	0.2883	0.2	0.03–1.6	0.1412	0.3	0.1–0.9	0.0267
Todd's paresis	0.5	0.1–2	0.3064	0.3	0.04–2.3	0.2504	—	—	—
Abnormal EEG	1.6	1–2.7	0.0572	2	1.1–3.5	0.0262	0.7	0.3–1.9	0.5186
Seizures while asleep	1.7	1.1–2.8	0.0166	1.9	1.1–3.3	0.0255	1.1	0.5–2.6	0.7967
Partial seizures	0.9	0.6–1.4	0.5720	0.7	0.4–1.3	0.2815	0.8	0.3–1.9	0.5850
Prior febrile seizures	1.1	0.6–1.9	0.6938	1.4	0.7–2.8	0.2682	0.5	0.1–1.7	0.2571
Prior neonatal seizures	2.0	0.7–5.5	0.1755	—	—	—	1	0.3–3.2	0.9299
Family history of seizures	1.2	0.6–2.4	0.5174	1.6	0.8–3.3	0.1884	—	—	—
First seizure treated	1.3	0.8–2.3	0.2837	1.5	0.7–3	0.2606	0.4	0.1–0.9	0.0357

Univariate analysis. Cox proportionate hazards model.
RR, rate ratio; 95% CI, 95% confidence interval (N = 133).

group, at 1 and 2 years, respectively. Stratification of the sample by etiology revealed different risk factors for patients with symptomatic and idiopathic/cryptogenic etiologies.

Idiopathic/cryptogenic etiology group

In this group, the statistically significant predictors of a higher recurrence risk were an abnormal first EEG and the occurrence of the first seizure during sleep, whereas a patient age of 3 to 10 years was a significant predictor of a lower recurrence risk (Table 4).

Age at first seizure: The recurrence risk was 27% and 36% for patients aged 3 to 10 years, against 51% and 59% for the rest, at 1 and 2 years, respectively (Fig. 3A).

First EEG: The Kaplan-Meier estimate of recurrence risk was 50% and 62% for children with an abnormal EEG, compared with 34% and 42% for children with a normal EEG, at 1 and 2 years, respectively (Fig. 3B).

Sleep state at time of first seizure: The recurrence risk was 52% and 65% for children whose first seizure occurred during sleep, in contrast to 32% and 39% for children whose seizure occurred while they were awake, at 1 and 2 years, respectively (Fig. 3C).

Symptomatic etiology group

In this group, a patient age of 3 to 10 years, treatment after the first seizure, and the occurrence of status epilepticus as the first seizure were significant predictors of a lower recurrence risk (Table 4). The recurrence risk was 56% and 89% for the children aged 3 to 10 years, compared with 88% and 100% for other patients, after 1 and 2 years, respectively. Although the difference is statistically significant, the clinical relevance of this finding is limited, because most of the patients, in both groups, finally recurred.

Although the recurrence risk was lower in the six patients with symptomatic etiology who presented with status epilepticus as their first seizure, we discount this finding because five of them were receiving antiepileptic treatment.

Multivariate analysis

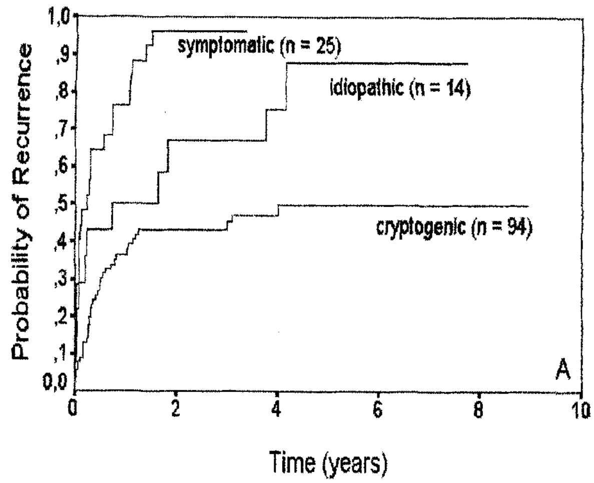
Multivariate analysis using the Cox proportional hazards model included the factors that were statistically significant in the univariate analyses. In the overall cohort, "symptomatic etiology" and "age 3 to 10 years" remained in the model. "First seizure during sleep" was almost statistically significant. In the idiopathic/cryptogenic group, statistically significant predictors of recurrence risk were "age 3 to 10 years," "abnormal first EEG," and "first seizure while asleep." In the symptomatic group, no factor remained significant. Table 5 shows the factors that were included in multivariate analysis.

DISCUSSION

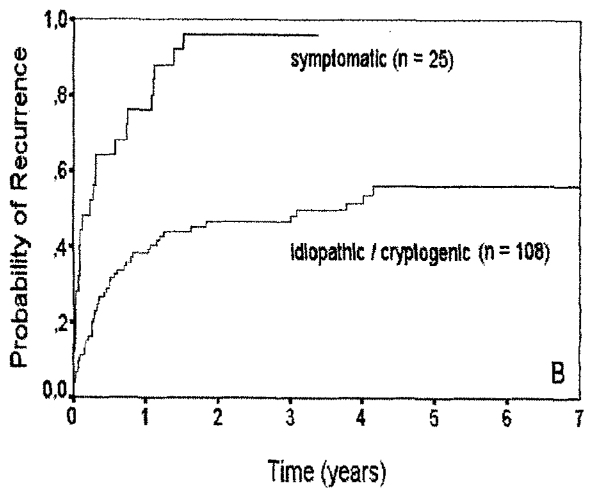
This study was designed to investigate the recurrence risk after a first unprovoked seizure in childhood. The choice of the 14-year age limit was determined by the characteristics of the organization of the National Health Service in Spain.

The most important predictor of recurrence risk in our study was the etiology of the seizures. The recurrence risk at 2 years was 46% for idiopathic/cryptogenic seizures and 96% for symptomatic seizures. The recurrence risk was different for idiopathic and cryptogenic seizures: 67% and 43%, respectively, at 2 years. This aspect has not been considered in previous studies.

Most studies of the recurrence risk in patients with some type of neurological abnormality (4,7,8,10–12,14,23,24) have found some increase in recurrence risk associated with such abnormalities. In the meta-analysis performed by Berg and Shinnar (15), the pooled recurrence risk at 2 years was 32% for patients with idiopathic/cryptogenic seizures and 57% for patients with symptomatic seizures. Nevertheless, it should be taken



at risk 133 40 23 10 3 0



at risk								
S	25	6	1	1	0			
I/C	108	55	39	34	23	16	10	6

FIG. 2. Risk of seizure recurrence after a first unprovoked seizure as a function of etiology. **A:** Considering three etiological groups. **B:** Considering two etiological groups. S, symptomatic; I/C, idiopathic/cryptogenic.

into account that the meta-analysis includes patients from all age groups, which could provoke some confusion, because the etiology of epilepsy changes significantly with age and the risk of recurrence after a first unprovoked epileptic seizure is not the same for all causes of symptomatic epileptic seizures. In fact, as occurs in our series, most symptomatic seizures suffered by patients aged less than 15 years are associated with neurological deficits presumably present at birth manifested by mental retardation and cerebral palsy (17). In two series, the recurrence risk in patients with such antecedents was 100% (14) and 92% (11) after 1 year, figures that were much higher than in patients with symptomatic seizures attributable to other causes. This circumstance explains the higher recurrence risk in the symptomatic group in our study.

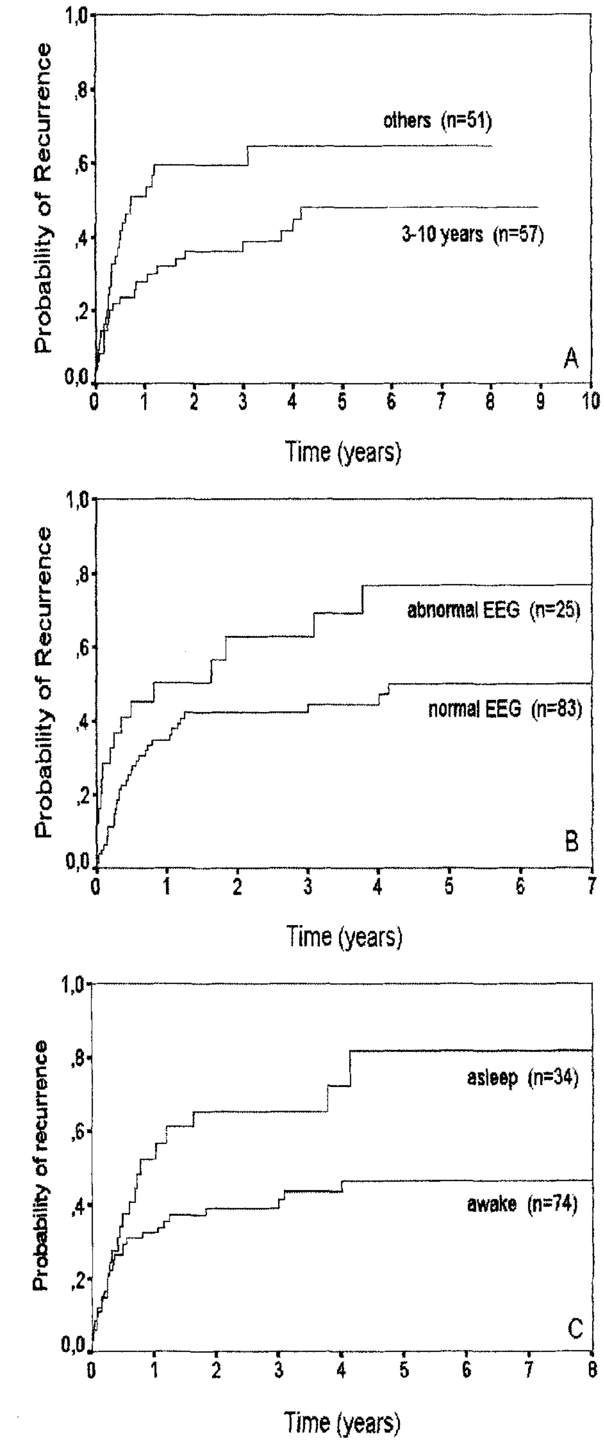


FIG. 3. Risk of seizure recurrence after a first unprovoked seizure in idiopathic/cryptogenic group. Kaplan-Meier curves. **A:** Patients 3–10 years old, compared with other patients (p = 0.0293). **B:** Abnormal EEG, compared with normal EEG (p = 0.0262). **C:** First seizure asleep, compared with first seizure awake (p = 0.0255).

TABLE 5. *Predictors of recurrence risk after a first unprovoked seizure*

	RR	95% CI	p
Overall cohort			
Symptomatic etiology	3.5	2.1–5.8	0.0000
Age 3–10 years	0.5	0.3–0.8	0.0036
Seizure while asleep	1.5	0.9–2.5	0.0711
Idiopathic/cryptogenic group			
Age 3–10 years	0.4	0.2–0.8	0.0047
EEG abnormal	2.5	1.3–4.7	0.0042
Seizure while asleep	2	1.1–3.5	0.0177
Symptomatic group			
Age 3–10 years	0.8	0.2–2.5	0.6996
Status epilepticus	0.5	0.1–2.1	0.3255
First seizure treated	0.6	0.2–1.7	0.2982

Multivariate analysis. Cox proportionate hazards model.
RR, rate ratio; 95% CI, 95% confidence interval (N = 133).

On the other hand, in the recent study of Shinnar et al. (8), with children aged less than 18 years, the recurrence risk in the symptomatic group was only 57% at 2 years. Again, a different etiological distribution could be the cause of the differences between this study and ours. Although the authors do not state the specific causes of the symptomatic seizures, it is noteworthy that whereas the two series are similar regarding the average age of the patients and the percentage of patients presenting symptomatic seizures, in the Shinnar et al. study only 21% of the patients with symptomatic seizures were 3 years old or younger, compared with 48% in our series. These same authors found that the recurrence risk among children aged less than 3 years with symptomatic seizures was almost 95% at 2 years. The authors suggested that this effect could be related to the fact that neurological deficits are more severe in this group, which is consistent with our discussion, because most of these patients probably had neurological handicaps at birth.

The predominance of patients aged less than 3 years among patients with symptomatic seizures, as found in our study, coincides with the results of previous population studies (25,26). For example, in the study by Ellenberg et al. (25), which included patients aged up to 7 years, 64% of the patients with unprovoked seizures associated with neurological deficits present at birth were aged less than 3 years.

Little is known about the influence of age on the recurrence risk, within the group of pediatric patients (4,8). Our results reveal an important decrease in recurrence risk for patients aged 3 to 10 years, especially in the group with idiopathic/cryptogenic seizures.

In our study, an abnormal EEG was an important predictor of recurrence risk, but only in the group of patients with idiopathic/cryptogenic seizures. Most studies have also found the same association (1,2,7,8,10–12,27).

The onset of the first seizure during sleep was another important predictor of recurrence in our study, but again only for the idiopathic/cryptogenic etiology group. This factor has been studied elsewhere only by Shinnar et al. (8,28), who also found the same fact.

In a recent study (8), multivariate analysis showed a greater recurrence risk for patients in the symptomatic group with a personal history of febrile convulsions. This effect was not observed in our study; indeed, we found quite the opposite tendency. In another two earlier studies (4,12), febrile convulsions were only associated with a slight, nonsignificant increase in the risk of recurrence. Other factors associated, although not consistently, with an increase in the recurrence risk include partial seizures, Todd's paresis, either multiple seizures or status epilepticus, a personal history of neonatal convulsions, and a family history of seizures (7,9–13,15,28). In our study, no significant association with recurrence risk was found for any of these factors.

Regarding the global recurrence risk, our results confirmed the findings of other authors (14,15) that it depends on the methodology used. In what Berg et al. (15) termed the first-seizure method, only cases in which the patient is attended after a first unprovoked seizure are included; patients with any history of unprovoked seizures are excluded. Patients with certain types of seizures that are characterized by a high recurrence rate, such as absences, myoclonic seizures, or infantile spasms, are, by definition, excluded. On the other hand, the recent-onset epilepsy method (15) includes all the cases attended, even if several seizures had previously occurred. It is intuitively apparent that the risk of recurrence will be higher when the second method is used. In our series, the recurrence risk after a first unprovoked seizure was 74% and 79% using the latter method and 57% and 64% using the first method, at 2 and 5 years, respectively. In a meta-analysis, Berg et al. (15) found a pooled recurrence risk of 42% at 2 years in studies using the first-seizure method and 67% in the studies using the recent-onset epilepsy method. In our series, the recurrence risk with the recent-onset epilepsy method is close to that obtained in the Berg et al. (15) meta-analysis, and is similar to the 75% recurrence risk after 2 years found in an important population-based study (14). The recurrence risk obtained when the first-seizure method is used is greater than that calculated in the meta-analysis, mainly due to the higher recurrence risk in the symptomatic group. Other ill-defined factors, such as the proportion of patients seen after a single seizure, could influence the recurrence risk. For example, in the Hauser et al. (10) study, 74% of patients looked for medical attention after having suffered more than one seizure and were thus excluded from the study. This percentage is much higher than the 39% in our study. Because the recurrence risk is greatest during the first few months after a seizure, if a significant percentage of patients are excluded because they have already suffered a recurrence, the recurrence risk obtained for the remaining patients is likely to be lower. Most authors, however, do not provide figures for this variable. This factor could have contributed to the slightly higher recurrence risk found in our series among the idiopathic/cryptogenic group.

Patients who were attended after a first seizure differed significantly from those who were attended after more than one seizure in several aspects: etiology, type of seizure, presentation as status epilepticus, frequency of EEG abnormalities, and history of febrile convulsions. Some of these factors are important predictors of recurrence risk.

There are disagreements among authors regarding the most appropriate way to study recurrence after a first seizure (14,15). Because the two populations are different, the exclusion of patients with previous seizures is not a simple methodological question. In truth, neither of the two methods is entirely satisfactory. First-seizure method provides information of greater practical interest, because it only makes clinical sense to examine the recurrence risk when the patient has suffered a single seizure. This, however, does not give an exact idea of the recurrence risk for the whole group of those affected by seizures, because it ignores a subgroup who, for various reasons (including the type of seizure suffered), attend the clinic only after they have suffered several seizures. The second method may more closely fit the population as a whole, but it is difficult to be sure that absolutely all patients with seizures in a population have been included, because some patients with single seizures, or even recurring seizures, never seek medical attention.

One limitation of our study is that, although standard practice of physicians in our catchment area is to remit patients with a possible diagnosis of epileptic seizure to our hospital for study, it is not possible to guarantee that all children with seizures in the population have been attended in the hospital. The characteristics of the sample obtained (29), especially with respect to the etiology of seizures, are similar to those found for the general population in the Rochester studies for children aged 0 to 15 years (17), which suggests there is not a significant bias. Regarding the recurrence risk after a first unprovoked seizure and the predictors of this risk, this study reproduces the situation as its takes place in the clinical practice when a patient comes to medical attention for the first time. This study is one of the few available about the recurrence risk after a first unprovoked seizure in childhood. Its main virtues are that it is prospective, uses survival analysis methodology, and follows ILAE recommendations for epidemiological studies.

The first conclusion derived from our study is that patients attended after a first seizure differ in various clinical aspects from those attended after two or more seizures, and thus the inclusion of the first or both types of patients is not a methodological question but, in fact, different populations are being studied. With respect to predictors of recurrence risk, our study highlights the importance of a symptomatic etiology, probably because most patients of this age have a congenital neurological deficit. In the symptomatic etiology group, the recurrence risk is so high that it is difficult to identify clinically relevant risk factors. Regarding the risk factors in the idiopathic/cryptogenic group, our findings confirm the role of the EEG as described by most authors and stress the importance of the patient's age and sleep state, which had previously received little attention.

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